

Topic	Description
Module	Drive Technology
Module Code	MECH-B-5-ANT-ANT-ILV
Semester	WS 2025
Lecturer	Daniel T. McGuiness, Ph.D
ECTS	5
SWS	3
Lecture Type	ILV
Coursework Name	Individual Assignment
Work	Individual
Suggested Private Study	10 hours
Submission Format	Submission via SAKAI
Submission Deadline	14.12.25 23.59
Late Submission	Not accepted
Resubmitting Opportunity	No re submission opportunity

No lecture time is exclusively devoted to the aforementioned assignment.

A portion of the mark for every assignment will be, where applicable, based on style. Style, in this context, refers to organisation, flow, sentence and paragraph structure, typographical accuracy, grammar, spelling, clarity of expression and use of correct IEEE style for citations and references. Students will find *The Elements of Style (3rd ed.)* (1979) by Strunk & White, published by Macmillan, useful with an alternative recommendation being *Economist Style Guide (12th ed.)* by Ann Wroe.

Submission Format and Requirements with L^AT_EX

Before submitting your work for assessment please follow the following requirements.

1. **The work must be done using L^AT_EX and no other format will be accepted.** You are allowed to use whatever software you wish to write (i.e., Overleaf, T_EXStudio, LyX, ...). The work must also conform to the **MCI Documentation guidelines** (which is in SAKAI). A template is provided to you [here](#) which was written by the author of this document.

If you are to use **mcidoc** template, set the `document-state` as `Report`.

To learn L^AT_EX, the author has tutorials which can be accessed [here](#).

Please treat this assignment as a great practice to learn L^AT_EX as you will need it when you are writing your thesis.

2. The report (and all its content) **must** be submitted as a `.pdf` along with all the necessary files (`.tex`, `.sty`) to compile it. Submit one (1) `.zip` file containing everything with the following name pattern.

`[STUDENT-ID]_[STUDENT-NAME]_[STUDENT-SURNAME].zip`

For example, a student named **Jason Brigham** with a student ID **2710704051** would send their submission as:

`2710704051_Jason_Brigham.zip`

and their submission would be in arranged as the following directory:

`2710704051_Jason_Brigham/`

```
├── tex/
│   ├── 2710704051_Jason_Brigham.tex, *.pdf, mcidoc.cls, custom.sty, ...
│   └── figures/
│       ├── .jpg, .png, ...
└── HW_Content/
    ├── src/, include/, .cpp, .bash, .py, ...
```

where `...` are any additional requirement (i.e., `.bib`, `.glo`) you may need for T_EX and the `HW_Content` is any content which is relevant to the Homework (i.e., if your HW is about python, this is where all the python scripts would be kept, or where your ROS 2 source files would be, or where your C++ codes would be for a project.)

3. The work must be submitted to SAKAI and **no link to personal repo will be accepted.**

To make sure your submission conform to given requirements, please test out zipping/un-zipping and running any code before submission. Any deviations from the requirements will incur penalties to the final grade.

[Q1] A Toroid Winding 30

A coil of insulated wire of 700 turns and of resistance $5\ \Omega$ is closely wound on iron ring. The ring has a mean diameter of 0,30 m and a uniform cross-sectional area of $800\ \text{mm}^2$. The diagram for it is shown below:

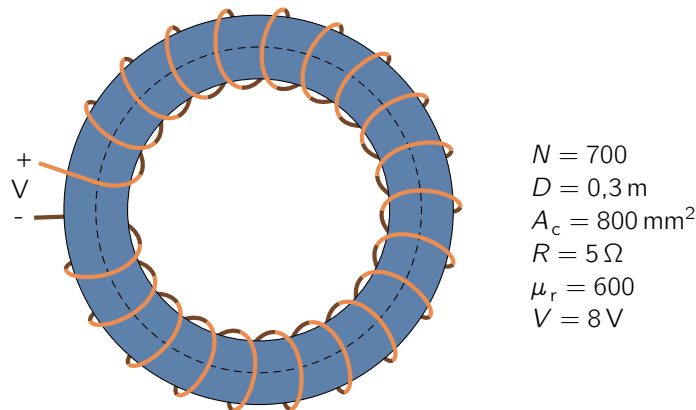


Figure 1: The wound toroid.

Based on the information given, please calculate the total flux in the ring when a DC supply of 8 V is applied to the ends of the winding.

Assume a relative permeability (μ_r) of 600.

[Q2] Transformer Power Losses 20

A 230 V, 50 Hz transformer has 300 primary turns. When you do measurements you notice it draws 6 A at 0.30 p.f. lagging at no-load. Based on this information, please determine:

- Core loss, (5)
- Magnetising current, (5)
- Core loss resistance and magnetising resistance. (10)

For this question you can neglect magnetic leakage, resistance of the winding and the primary no-load current in relation to the full load current.

[Q3] Armature Reaction 20

Please answer the questions briefly (max 200 words per question).

- What is armature reaction in context of a DC drive?, (5)
- What are its unwanted effects?, (5)
- Explain the disadvantages of armature reactions. (5)
- How can it be minimised? (5)

[Q4] Induction Machine Calculations 30

A three-phase delta (Δ)-connected 230 V, 8,0 kW 50 Hz four-pole induction machine has the following parameter values referred to the stator:

$R_1 = 0,284 \Omega$	$R_2 = 0,196 \Omega$	$X_1 = 0,473 \Omega$	$X_2 = 0,189 \Omega$	$X_m = 15,01 \Omega$
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The total friction, windage, and core losses are assumed to be constant at 373 W, independent of the load. For a slip of 2.3 percent, calculate:

- i. The speed, (6)
- ii. The stator current, (6)
- iii. The output torque and power (6)
- iv. The power factor, and (6)
- v. The efficiency. (6)

when the motor is operated at rated voltage and frequency.

Report You are to write a report on your individual assignment which should explain the calculation used in solving the given questions. A template structure for this assignment could be as follows. Per each question:

- An introduction to the question, and the method used to tackle.
- Explanation (i.e, conceptual and mathematical) of the method(s) used in the tackling of the problem. Calculations should be explicit and detailed with proper units and calculations clearly written.

Before submission, please read **Submission Format and Requirements with L^AT_EX**.